

Tunneling

Let $V(x)$ be the barrier potential

$$V(x) = \begin{cases} 0 & x < -a \\ V_0 & -a \leq x \leq a \\ 0 & x > a \end{cases}$$

Find tunneling probability T for a particle with mass m and energy $E < V_0$.

§1

We have the following Schrodinger equations, one for each region of $V(x)$.

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi_1 &= E\psi_1 & x \leq -a \\ -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi_2 + V_0\psi_2 &= E\psi_2 & -a \leq x \leq a \\ -\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi_3 &= E\psi_3 & x \geq a \end{aligned}$$

The solutions are

$$\begin{aligned} \psi_1(x) &= A \exp(ikx) + B \exp(-ikx) \\ \psi_2(x) &= C \exp(i\beta x) + D \exp(-i\beta x) \\ \psi_3(x) &= F \exp(ikx) \end{aligned}$$

where

$$k = \frac{\sqrt{2mE}}{\hbar} \quad \beta = \frac{\sqrt{2m(E - V_0)}}{\hbar}$$

Boundary conditions at $x = -a$.

$$\begin{aligned} \psi_1(-a) &= \psi_2(-a) \\ \psi_1'(-a) &= \psi_2'(-a) \end{aligned}$$

Boundary conditions at $x = a$.

$$\begin{aligned} \psi_2(a) &= \psi_3(a) \\ \psi_2'(a) &= \psi_3'(a) \end{aligned}$$

From boundary conditions for ψ_2 and ψ_3 we have

$$C \exp(i\beta a) + D \exp(-i\beta a) = F \exp(ikx) \quad (1)$$

and

$$i\beta C \exp(i\beta a) - i\beta D \exp(-i\beta a) = ikF \exp(ikx) \quad (2)$$

Add $i\beta$ times (1) to (2) to obtain

$$2i\beta C \exp(i\beta a) = (i\beta + ik)F \exp(ika)$$

Hence

$$C = \frac{(\beta + k)F \exp(ika - i\beta a)}{2\beta}$$

Add minus $i\beta$ times (1) to (2) to obtain

$$-2i\beta D \exp(-i\beta a) = (-i\beta + ik)F \exp(ika)$$

Hence

$$D = \frac{(\beta - k)F \exp(ika + i\beta a)}{2\beta}$$

From boundary conditions for ψ_1 and ψ_2 we have

$$A \exp(-ika) + B \exp(ika) = C \exp(-i\beta a) + D \exp(i\beta a) \quad (3)$$

and

$$ikA \exp(-ika) - ikB \exp(ika) = i\beta C \exp(-i\beta a) - i\beta D \exp(i\beta a) \quad (4)$$

Add ik times (3) to (4) to obtain

$$2ikA \exp(-ika) = (ik + i\beta)C \exp(-i\beta a) + (ik - i\beta)D \exp(i\beta a)$$

Hence

$$A = \frac{(k + \beta)C \exp(ika - i\beta a)}{2k} + \frac{(k - \beta)D \exp(ika + i\beta a)}{2k}$$

Substitute for C and D .

$$A = \frac{(k + \beta)^2 F \exp[2ia(k - \beta)]}{4k\beta} - \frac{(k - \beta)^2 F \exp[2ia(k + \beta)]}{4k\beta}$$

Hence

$$\frac{A}{F} = \frac{(k + \beta)^2 \exp[2ia(k - \beta)]}{4k\beta} - \frac{(k - \beta)^2 \exp[2ia(k + \beta)]}{4k\beta} \quad (5)$$

§2

Substituting for k and β we have for tunneling probability T

$$T^{-1} = \frac{A}{F} \left(\frac{A}{F} \right)^* = 1 + \frac{1}{8} \left(\frac{E}{V_0 - E} + \frac{V_0}{E} + 1 \right) \left[\cos \left(\frac{4a}{\hbar} \sqrt{2m(E - V_0)} \right) - 1 \right] \quad (6)$$

Transform $\cos$ to $\sinh$.

$$T^{-1} = 1 + \frac{1}{4} \left(\frac{E}{V_0 - E} + \frac{V_0}{E} + 1 \right) \sinh^2 \left(\frac{2ia}{\hbar} \sqrt{2m(E - V_0)} \right) \quad (7)$$

Cancel the imaginary unit by swapping E and V_0 .

$$T^{-1} = 1 + \frac{1}{4} \left(\frac{E}{V_0 - E} + \frac{V_0}{E} + 1 \right) \sinh^2 \left(\frac{2a}{\hbar} \sqrt{2m(V_0 - E)} \right)$$

Equivalently

$$T^{-1} = 1 + \frac{V_0^2}{4E(V_0 - E)} \sinh^2 \left(\frac{2a}{\hbar} \sqrt{2m(V_0 - E)} \right) \quad (8)$$

§3

Check dimensions.

$$\frac{2a}{\hbar} \sqrt{2m(V_0 - E)} = \frac{[\text{m}]}{[\text{J s}]} \sqrt{[\text{kg}] [\text{J}]} = [1] \quad (9)$$

§4

Eigenmath script